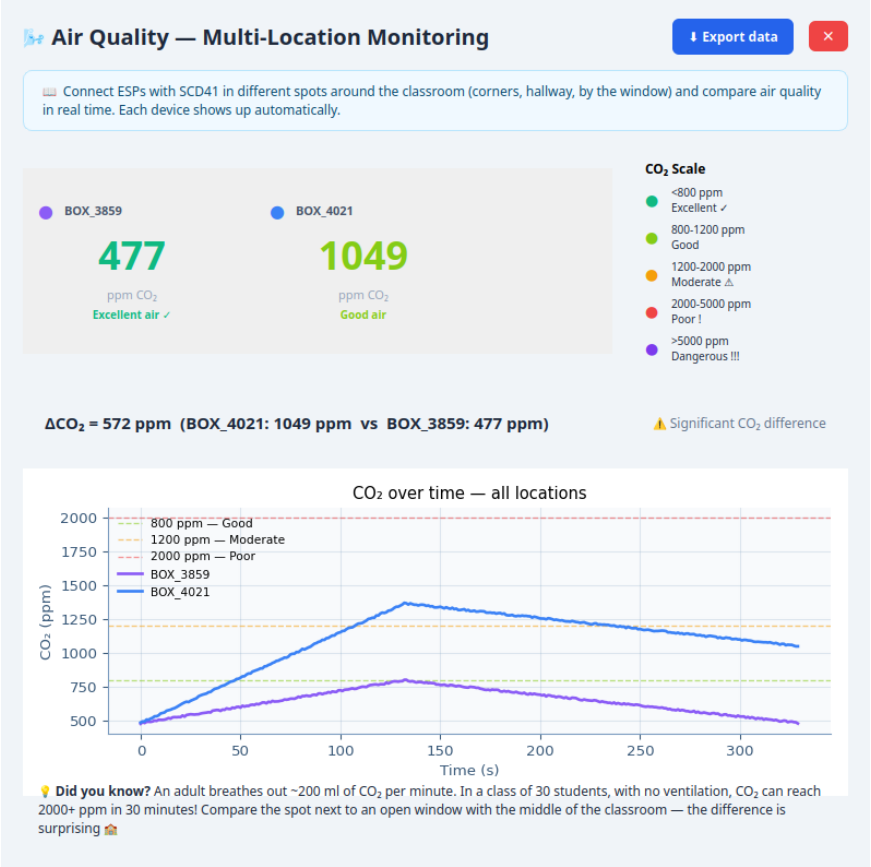
# Air Quality

Several boxes at once — and the threshold below which a $CO_2$ difference means nothing

**Level:** Explorer and Scientist — ages 11–19 (Romanian school years V–XII) • **Time in class:** 20 min demonstration, 50 min investigation, or a two-week project • **Sensors:** SCD41 on two or more boxes




*The activity screen. The values in the picture are simulated; your own will differ.*

## Before you start

**Who it is for**

Explorer and Scientist — ages 11–19 (Romanian school years V–XII). The activity card carries the **Explorer** badge and reads **Grades V–XII · Ages 11-19 · several boxes**; as a projected demonstration it goes lower still: a year IV class can watch two gauges and understand which corner needs air. **As an investigation**, though, it requires the idea of a *detection threshold*, so from year V upwards. The most direct entries are in **year VII** (biology: respiration) and **years IX–XII** (research method, environmental projects).

**Assumed knowledge**

**The guided activity** *Indoor Pollution*, done first: the protocol and the idea of a rate of rise come from there. In addition: the difference of two values and the arithmetic mean. For the project: an $XY$ chart in a spreadsheet.

| | |
|---|---|
| **Time in class** | Short: 20 min (demonstration). Full: 50 min (one run). Project: two weeks at 10–15 min a day, plus two analysis lessons. |
| **Teacher preparation** | 30 min — **the co-location test of every box**, at least ten minutes on the same table at the same height; labelling each box with the exact name it shows on screen; choosing and marking the locations; checking WiFi coverage at each; and printing the room plan. **Give each box a name** in **Settings**, under **Box names** (for example "Window", "Corner", "Door"): the box's name and colour then appear the same everywhere. |
| **Sensor settling time** | **Ten minutes** on first placing a box in a new spot, and **at least five** on every later move. The SCD41 gives a value about every five seconds, but the gas has to diffuse into the sensor's chamber, which takes far longer than one reading. **A box moved and read after a minute is still showing the air of where it was before.** |
| **Sensors and boxes** | SCD41 on **at least two boxes**. With only one, the summary line shows no number but the message **$\Delta CO_2$ = — \| Connect at least two boxes**. Four to six boxes make the investigation far richer, because they allow every location to be measured *simultaneously*. Every box needs firmware with $CO_2$: 2.2.0 or later. A box still on the old USB-only firmware (`serial_only`) is recognised, and the app says to flash the WiFi firmware, which works over USB too (`firmware` folder, flashing guide). |

**Three routes**

| | |
|---|---|
| **Short** | 20 min: projected demonstration — two boxes side by side on the same table, then one by the window and one in the furthest corner, and the discussion of what the gap between the two curves means. |
| **Full** | 50 min: a single run — co-location, four locations, the build-up and purge experiment, export and graph. |
| **Project** | Two weeks: the same locations at the same times, with a duty pair; a ventilation policy tested in the second week; and a short report for the school leadership. |

**The one-box activity first, then the many-box app**

***Indoor Pollution* is done first.** It is not a smaller version of this one but the lesson in protocol this app needs in order to mean anything: the baseline taken in an aired room, the time recorded alongside the value, the number of people and the state of the window written down every time, and the idea that what matters is the *rate*, not the value.

**What the app adds** is simultaneity: several boxes read *at the same moment*, so the time of day can no longer explain the difference between the corner and the window. And with it, the app adds the thing a single box cannot do: **measuring the detection threshold directly**, by putting the boxes side by side before spreading them out.

A class handed six boxes before it has ever held one still for five minutes produces six incomparable datasets. If you have time for only one, do the guided activity: it has a complete conclusion without the app. The app without it does not.

## 1. What the app shows and what it does not

Unlike the guided activities, this is not a step-by-step wizard. It is an **open instrument** with a single purpose: the simultaneous comparison of several locations. There is no green button, no **Back**, no **Finish** and no observation journal. **You set the pace of the lesson**, and the investigations in section 6 are ready to run as written.

- **One gauge per connected box**, appearing automatically, labelled with the box's name (the one given in **Settings**) and in **its colour**; the gauge also shows the $CO_2$ band it is in.

- **A single comparison chart**, with one line per box, in **the same colour as on its gauge and on the Live Data page**; the $CO_2$ bands are drawn as a coloured background with their names, and time is read in minutes and seconds.
- **The $\Delta CO_2$ line**, which shows the difference between the highest and the lowest box, names them both, and adds an automatic verdict in words underneath.
- **The $CO_2$ scale**, printed on screen next to the gauges, with its five bands and their ranges.
- **A tip at the bottom**, *Did you know?*, with the order of magnitude of the build-up in a closed room.
- **The Export data button**, which writes one dated file per connected box.

**Technical note — what is on screen, and what the automatic verdict does not tell you**

**How it starts.** From the **Activities** page, from the **Air Quality** card (in the tools section, marked **several boxes**). Because the app works with several boxes at once, **no box-selection dialogue appears**: it automatically takes *every* connected box that reports $CO_2$, and each appears on screen by itself within seconds. While there are none, it reads **No box connected: switch on a box with a $CO_2$ sensor**. With several boxes, name them in **Settings → Box names** before the lesson; otherwise their technical identifiers (`BOX_…`) appear on screen.

**The $\Delta CO_2$ line.** With at most one box connected it reads **$\Delta CO_2$ = — | Connect at least two boxes**. From two boxes upwards it shows the form **$\Delta CO_2$ = 480 ppm (Corner: 1520 ppm vs Window: 1040 ppm)**, so both the difference and the two ends, with the box names.

**The automatic verdict says two things, not one.** Under the Δ line come two separate lines, **Level:** and **Difference:**, because they answer different questions. The level is read from the highest value, at the same thresholds as the *Moderate* and *Poor* bands: from 2000 ppm, *"Corner at 2150 ppm: open the windows now"*; from 1200 ppm, *"…: the air is getting stale"*; below 1200, *"highest is … ppm: all places well ventilated"*. The **Difference:** line appears only from two boxes upwards.

**Until recently the verdict was decided on the difference alone**, at 300 and 600 ppm — which had an ugly consequence: a whole floor sitting at 2000 ppm has small differences between rooms, so the app printed *similar air quality in all locations* over a building that badly needed airing. If a colleague remembers that wording, it is worth saying it has changed.

**A small difference is no longer declared "similar".** From 600 ppm the line names the places ("Corner is 620 ppm above Window"); between 300 and 600 ppm it only says the places differ by that much. Below 300 ppm the app states how far apart the places are and **tells you to put the boxes side by side** before calling that difference real, in two steps: above 180 ppm it adds *"check it with the boxes side by side"*; at 180 ppm or less, *"Put the boxes side by side first"* — exactly the protocol of investigation 1. The reason is the one in the technical manual: the SCD41 has an accuracy of ±(40 ppm + 5% of reading), so about ±90 ppm per box at 1000 ppm, and **two boxes side by side can legitimately differ by some** 180 **ppm**. **Do the co-location first** — only it tells you the scatter of *your* boxes.

**The $CO_2$ scale.** On the gauges and as background bands on the chart: below 800 ppm **Excellent**, 800–1200 **Good**, 1200–2000 **Moderate**, 2000–5000 **Poor**, above 5000 **Dangerous**. The bands belong to the *level*; the line colours belong to the *boxes* — the two are kept apart, so that no box looks "dangerous" merely because of its colour. The WHO has no guideline value for indoor $CO_2$; the bands are a rounded version of the ventilation categories of the standard EN 16798-1 (see the scientific background); the tooltip that appears when you hover over the scale says so too.

**The chart.** A single set of axes, **$CO_2$ in every place**, with one line per box, each in its box's colour; a disconnected box is drawn dashed and marked **(offline)**, and the figure on

its gauge turns grey. Above the chart sit **Smooth**, **Pause** and **Clear**. For the projector, **Settings → Projector mode** enlarges text and lines. The chart holds the last four hours of readings, enough for a whole morning.
**How long you wait. Ten minutes** on first placing a box in a new spot and **at least five** on every move. The reading is taken at the end of the interval, not on arrival.
**Exporting. ↓ Export data**, top right. Because the app works with several boxes, it first asks for a *folder* — the dialogue is called **Choose the export folder** — and then writes **one separate file per box**, named `<box name>_YYYY-MM-DD.xlsx` (for example `Corner_2026-10-05.xlsx`; a box that has since disconnected still gets its file, with what it measured). On success **Export complete** appears with the number of files written; if a file cannot be written (for example because it is open in Excel), the app says so instead of staying silent. **Put each day's folder inside a folder named for that date**, or the second day overwrites the first.

## 2. Where this fits in the curriculum

- **Biology** — in **year VII**, human respiration: every pupil is a $CO_2$ source, and here it becomes visible how exhaled air distributes itself around the room, not merely that it accumulates.
- **Mathematics** — in **year VI**, the arithmetic mean and the difference; in **years VII–VIII**, the idea that a difference must be compared with *the scatter of the measurement* before it can be declared real. This is the most valuable statistical idea a lower-secondary pupil can learn, and here they can *measure* it.
- **Geography and environmental science** — in **year V**, the atmosphere; in **years IX–XII**, indoor environmental quality as a public health problem, and survey method.
- **Physics** — in **year VIII**, heat transfer by convection and air currents: why the corner by the door has different air from the corner by the window, in the same room.
- **Personal development** — the project ends with **a report for the school leadership**, supported by the pupils' own data.

The core skill is **telling a real difference from instrument noise**. The app reports the difference as such from 300 ppm and sends you to co-location below it; the pupils have to find out for themselves, by co-location, whether 300 ppm means anything for *their* boxes.

## 3. Time plan

| Route | Time | What happens |
|---|---|---|
| **Demonstration** | **20 min** | The screen projected. Two boxes side by side on the same table for ten minutes — *that is the co-location* — then one by the window and one in the furthest corner. Discussion of what the gap between the two curves means. |
| **Full** | **50 min** | A single run: co-location (10), spreading the boxes to four marked locations (5), the build-up with the class inside (15), the purge with the window open (10), export, graph and discussion (10). |
| **Project** | **2 weeks** | The same locations at the same times, 10–15 min a day, with a duty pair; in the second week a *ventilation policy* is tested; at the end, two analysis lessons and a short report for the school leadership. |

## 4. Materials

| Item | Per class | Notes |
|---|---|---|
| MagicBox units with SCD41 | 2–6 | **Two minimum**; with one the Δ line shows nothing. Four to six make the investigation far richer. |
| Laptop with the app | 1 | Every box connects to the same laptop, over WiFi or USB; their names are set once, in **Settings**. |
| Labels for the boxes | 1 per box | **Written with the name given to the box in Settings, in its on-screen colour** — otherwise nobody can tell which curve is which. |
| Labels for the locations | 4–6 | Mark the spots physically: the far corner, by the window, by the door, the middle of the room, the corridor. |
| Printed room plan | 1 per pupil | For the $\Delta CO_2$ map. |
| Extension leads or power banks | 2–6 | For boxes standing where there is no socket. |
| Paper log | 1 | Time, number of people, state of the window and the door. The same columns as in *Indoor Pollution*. |
| Shared folder or USB stick | 1 | All the dated files in one place, one folder per day. |

**Preparation. Do the co-location test before the lesson**, not during it: every box on the same table at the same height, out of draughts and away from anybody's breathing, for at least ten minutes. Record the scatter — the difference between the highest and lowest value — because it is the threshold below which nothing can be declared real. Label each box with its on-screen name (names are set in **Settings** → **Box names**). Choose and mark the locations, check the WiFi coverage at each, and print the room plan.

## 5. Scientific background for the teacher

### Why a room does not have one $CO_2$ value

The air in a room is not an instantaneously uniform mixture. The sources — people — are unevenly distributed, and fresh air enters at one place only: the window or the door. A field of concentrations establishes itself between them.

The main mechanism is not molecular diffusion, which is far too slow — the diffusion coefficient of $CO_2$ in air is of the order of $1.6 \times 10^{-5}$ m$^2$/s, so crossing a room would take hours — but **convection and air currents**: body heat, the radiator, the door being opened. That is why the corner furthest from the window and nearest a group of pupils has, repeatably, the highest value.

The typical difference between the best and worst spot in a full room with the window shut is 100–400 **ppm**; with the window open it *increases* at first — because the spot by the window drops sharply while the far corner lags — and only then narrows. **That moment, when ventilation temporarily increases $\Delta CO_2$, is the most counter-intuitive result of the activity** and is worth hunting for deliberately.

**How fast a room fills.** A seated pupil exhales about 0.15–0.2 L of $CO_2$ a minute. Thirty pupils in a 170 m$^3$ room therefore add about 30 ppm a minute, that is **roughly +1000 ppm in half an hour** with the windows shut — from 500 to about 1500 ppm, not the "2000 and more" often quoted. Smaller rooms, more pupils or a whole hour take it higher, of course.

## 5.1 The detection threshold: why co-location is the first investigation

The technical manual gives the SCD41 an accuracy of ±(40 ppm + 5% of reading). At 1000 ppm that is about ±90 ppm per box. Two perfectly working boxes placed side by side can therefore differ by **almost 200 ppm** without either being wrong.

The practical consequence is simple and severe: **a difference between two locations means something only if it is larger than the measured scatter of the boxes**. And that scatter is not read from a datasheet, it is *measured*, by standing the boxes side by side for ten minutes before spreading them out. Part of the scatter comes from the sensor's automatic self-calibration: each SCD41 recalibrates itself, assuming the lowest value it has seen over a few days is outdoor air. A box that has not seen fresh air for a long time (or has spent time in jars with plants) can be offset; check it outdoors, where it should read 420–450 ppm, and recalibrate it if needed (technical manual).

This is also the difference between what the app can say and what the class can. The app reports the difference as such from 300 ppm, with a fixed threshold valid for any set of boxes. The class can do better: it can say *here, with our boxes, a difference below ____ ppm cannot be told from noise*. A pupil who can frame that has understood more scientific method than any textbook chapter conveys.

## 5.2 The air change rate, measured from your own purge curve

When the window opens, every box sees an exponential decay towards the outdoor value:

$$c(t) = c_{\text{out}} + \left(c_0 - c_{\text{out}}\right) \mathrm{e}^{-t/\tau}.$$

The constant $\tau$ can be read straight off the data: it is the time in which the excess over outdoors falls to about 37% of its initial value, or — more easily for pupils — **the half-time divided by** 0.69.

With several boxes you obtain a $\tau$ for each location, and that is the most useful information in the whole project: **not all of the room airs at the same speed**. The corner with the largest $\tau$ is the one that needs intervention — another window, an open door, or simply moving the desks.

## 5.3 Why $CO_2$ tracks infection risk and concentration

$CO_2$ is not itself dangerous at classroom levels, and it does not make breathing "harder". **The World Health Organization has no guideline value for indoor $CO_2$**; the European reference is the ventilation standard **EN 16798-1**, which grades buildings by how far indoor $CO_2$ exceeds outdoor: about +550 ppm (category I), +800 ppm (category II) and +1350 ppm (category III), roughly 970, 1220 and 1770 ppm with 420 ppm outdoors. $CO_2$ is a **ventilation marker** — an **indicator of the rebreathed fraction**: if the room's air is at 1200 ppm and outdoors is 420, then the excess of 780 ppm comes entirely from lungs, and its ratio to the concentration of exhaled breath (40 000 ppm) gives the fraction of air that has been breathed before — in this case about 2%.

That same fraction carries aerosols, which makes $CO_2$ a useful, though indirect, indicator of airborne transmission risk. The link with attention is less certain: some studies find poorer performance in badly ventilated rooms, others do not; most likely it is the stale air as a whole (warmth, moisture, other substances) that matters, and $CO_2$ merely signals it.

**Be exact about the limits.** The sensor does not see dust, smoke, volatile organic compounds or damp. **Conclusions about "pollution" in the broad sense need other instruments**, and the pupils' report has to say so.

# 6. Ready-made investigations

## 6.1 Investigation 1: co-location — first, always

**Research question.** How much do our boxes differ from each other when they are measuring the *same* air?

**Method.** Every box on the same table, at the same height, at least a metre from any person, out of draughts and out of the sun. Leave them **ten minutes**. Read $\Delta CO_2$ from the screen every minute over the last five.

**What to measure.** The largest value of $\Delta CO_2$ over those five minutes. **That is the class's co-location scatter.**

**Expected finding.** Typically 30–150 ppm, depending on the boxes and the absolute value. Pupils then write the sentence that governs everything else: *we will treat a difference between two locations as real only if it exceeds _____ ppm.* **Without this investigation, every other one produces numbers nobody can interpret.**

## 6.2 Investigation 2: the $\Delta CO_2$ map of one room

**Research question.** Does our room have one air quality?

**Method.** The boxes are placed simultaneously in four to six marked locations: the corner furthest from the window, by the window, by the door, the middle of the room, perhaps the corridor. Leave **ten minutes** on first placing. Read each gauge and record it on the printed room plan.

**What to measure.** $CO_2$ at every location, simultaneously; $\Delta CO_2$ between the largest and the smallest; **and the comparison of that $\Delta$ with the co-location scatter**.

**Expected finding.** In a full room with the window shut, a $\Delta$ of 100–400 ppm. The verdict is given in one of three forms: *detected* if $\Delta$ clearly exceeds the scatter, *not detected* if it is below it, and *borderline* if it is comparable. **All three are correct answers**, and the third is the most honest thing a researcher can say.

## 6.3 Investigation 3: build-up and purge — the air change rate

**Research question.** How fast does each corner of the room air out?

**Method.** With the class inside and the window shut, let the $CO_2$ rise for 15–20 minutes, recording every five. Then open the window — and the door if you can — and record every minute for ten minutes.

**What to measure.** For each box: the rate of rise in ppm/min; the half-time of the excess over outdoors; and from it, $\tau$.

**Expected finding. The rates of rise differ little between locations, but the purge rates differ a great deal**: the spot by the window empties several times faster than the opposite corner. And at the start of the purge, $\Delta CO_2$ *rises* — the result that surprises the whole class.

## 6.4 Investigation 4: does the timetable show up in the data?

**Research question.** Can we read the school's timetable from the $CO_2$ curve?

**Method.** One box left logging for a whole day in a fixed spot, with the paper log filled in by pupils: the time, how many are in the room, the state of the window. Export at the end of the day.

**What to measure.** The day's curve, with every peak and dip matched to an entry in the log.

**Expected finding.** Breaks appear as valleys, lessons as climbs, and lunch as a deep fall. **The worst hour of the day can be named exactly**, and that is the figure that goes into the report.

### 6.5 Investigation 5: testing a ventilation policy

**Research question.** Does the rule we proposed actually work?

**Method.** In the first week, measure with nothing changed. In the second, apply one specific rule — for example *the window wide open for five minutes at every break* — and measure identically. Nothing else may change: same locations, same times, same boxes.

**What to measure.** The day's peak value and the fraction of time spent above 1200 ppm, for each week.

**Expected finding.** A clear fall in the fraction of time above the threshold, even if the peaks stay similar. **Comparing fractions of time rather than peaks is what makes the report convincing**, and it is exactly how a real intervention is evaluated.

**The investigation, in one table**

**The question.** Is the corner furthest from the window really less well ventilated than the spot beside it?
**The hypothesis.** Written **beforehand**: the $\Delta CO_2$ between the two locations will be _____ ppm, plus a reason. Not changed afterwards.
**What I change.** One thing only: the **location** of the box.
**What I measure.** $CO_2$ at each location, *simultaneously*, and the $\Delta CO_2$ from the screen.
**What I keep the same.** The same height at every location; the same boxes, with their labels; **a one-metre exclusion circle** around each box; the time recorded; and the state of the window and door unchanged for as long as the measurement lasts.
**How many repeats.** The measurement is repeated at three times of day — morning, midday, afternoon — and, if possible, with the boxes **swapped between locations** on the second run. If the difference stays with the *place* and not with the *box*, it is real.
**What makes a reading valid — and what makes it an artefact.** Valid is a reading taken after ten minutes at the first location and five at the others, with nobody closer than a metre, and **only if $\Delta$ exceeds the co-location scatter**. An artefact is anything else: a pupil sitting beside a box raises that box alone by hundreds of ppm; a box moved and read after a minute shows the air of where it was before; and a $\Delta$ smaller than the scatter does not mean "the places are the same" but **"we cannot tell"**, which is not the same statement.

## 7. Running a two-week project

### 7.1 The schedule

| When | Who | What happens |
|---|---|---|
| Day 0 | Teacher | Co-location of every box, labelling, choosing and marking the locations, checking WiFi. |
| Week 1, day 1 | Whole class, 50 min | Investigations 1 and 2. The class's detection threshold is established. Exporting is practised. |
| Week 1, daily | Duty pair, 10–15 min | Readings at 9, 12 and 15 at the fixed locations, plus one export a day. The paper log filled in every time. |
| Week 1, one day | Duty pair | A whole day of logging (investigation 4). |
| Week 2, daily | Duty pair | Identical, but with **the ventilation policy** applied (investigation 5). |
| Week 2, end | Whole class, 2 h | The files are gathered, the two weeks compared on *fraction of time above threshold*, and the report written. |

### 7.2 Keeping days comparable

- **Boxes are not moved between locations** during a week — or, if they are swapped, *all* of them are swapped, deliberately, as a test.
- **The same clock times.** A reading at 12:00 and one at 12:40 are not comparable in a room that is filling up.
- **The paper log is not optional.** Without the number of people and the state of the window, a week of data cannot be interpreted.
- **One folder per day.** The export writes dated files, but the box names repeat; without separate folders the second day overwrites the first.
- **Repeat the co-location at the start of week two.** If the scatter has changed, the threshold changes with it.

## 8. Analysis after the activity

## 8.1 What the data typically looks like

| Situation | Typical value | What it means |
|---|---|---|
| Co-location, all boxes together | $\Delta = 30$–$150$ ppm | **The class's detection threshold.** |
| Empty, aired room | 450–700 ppm everywhere | Small $\Delta$; nothing to detect. |
| Full room, window shut | $\Delta = 100$–$400$ ppm | Far corner highest, window lowest. |
| **Just after the window opens** | $\Delta$ *rises* to 400–800 ppm | The counter-intuitive result: the window drops sharply, the corner lags. |
| Ten minutes after airing | $\Delta$ falls below its previous value | The room has evened out. |
| Corridor against classroom | 300–800 ppm lower | Greater volume, fewer people. |
| $\tau$, by the window | 3–8 min | Empties fast. |
| $\tau$, far corner | 10–25 min | The spot that needs intervention. |

## 8.2 From $\Delta$ to a conclusion, step by step

1. **Write the co-location threshold** at the head of the page, before anything else. Say 80 ppm.
2. **Compute $\Delta$ for every pair of locations** — not just the maximum shown on screen, but every pair you care about.
3. **Compare each $\Delta$ with the threshold.** Above it, the difference is real. Below it, **it cannot be told apart** — and that is written exactly so, not as "the places are the same".
4. **Name the worst location** and say by how much it exceeds the threshold. "The corner by the cupboard is 240 ppm above the spot by the window, which is three times our 80 ppm threshold."

## 8.3 Questions to ask afterwards

1. The app said "the places differ by 310 ppm". Your boxes, side by side, differed by 120 ppm. What should the app say to be correct for *you*?
2. When we opened the window, $\Delta CO_2$ *rose*. How can ventilation make the room *less* uniform?
3. You swapped the boxes between locations and the difference stayed with the same place. What does that rule out?
4. The worst spot had a $\tau$ three times the best one's. What would you recommend, and what would it cost?
5. The room stayed below 2000 ppm all day. Can you say the air was clean? What exactly does this sensor not see?

## 8.4 Assessment ideas

- **Low stakes:** the pupil completes the room map with the simultaneous values and circles the worst location.

- **Standard:** the pupil compares every Δ with the co-location threshold and writes one of three verdicts for each pair — detected, not detected, borderline.
- **Extension: the one-page report for the school leadership** — the problem, two pieces of evidence, the policy tested, the result measured as *fraction of time above threshold*, and a stated limitation of the measurement.

### Common pitfalls and misconceptions

**Skipping the co-location.** Without it, no Δ can be interpreted and the class does nothing but repeat the app's verdict. It is the first investigation and it is not skipped.
**Taking the automatic verdict at face value.** The level threshold is sound, but the *difference* between rooms is still judged against the scatter you measured, not against a general threshold. The app itself tells you to put the boxes side by side; do it.
**Reading on arrival.** Ten minutes at the first location, five at every move. A box read after a minute shows the air of where it was before.
**Pupils sitting beside a box.** They raise that box alone by hundreds of ppm and invent a Δ. A one-metre exclusion circle, enforced.
**Missing labels.** With four curves on the chart and no labels on the boxes, nobody knows which line is which after five minutes.
**One export folder for every day.** The files carry the box name and the date, but a second day in the same folder invites confusion; one folder per day settles it.
**"Below the threshold means the same."** No. It means *we cannot tell*. The difference between those two statements is the methodological lesson.
**Conclusions about "pollution".** The sensor sees $CO_2$, which is ventilation. It does not see dust, smoke or volatile organic compounds. The report has to say so.

### Safety and classroom management

**Windows are opened by the teacher**, not by pupils, and nobody climbs on a sill or a radiator.
**Cold and draughts.** Brief, full airing is both more effective and less unpleasant than long, weak airing. Move pupils away from the window while it is open, and wedge the door so that it cannot slam.
**Boxes left in fixed locations** are announced to colleagues who use the room and labelled with the class and teacher. They do not stand in escape routes, on sills they can fall from, or where they can be knocked.
**Cables.** With four boxes powered in four corners, cables quickly become a problem. Use power banks where you can, and tape down what you cannot avoid.
**The corridor.** A box left in a corridor is announced to the school office.

### Adaptations for pupils with special educational needs

**The drawn map instead of the table.** The pupil is given the printed room plan and writes the value beside each location, then colours it: green below 800, yellow below 1200, red above. **The coloured map is a complete representation of the result** and requires no arithmetic.
**The threshold, written once and large.** For pupils who lose the thread among numbers, the co-location threshold is written once, large, at the head of the sheet, and every comparison refers to it. The question becomes binary: larger or smaller than the threshold?
**Three real roles for pupils with motor difficulties.** The activity involves walking about, but the roles that matter can be done from a desk:

- **the threshold keeper** — remembers the co-location scatter and has the right to reject any conclusion based on a Δ smaller than it; this is the role with the greatest scientific

value in the class;
- **the screen reader** — reads the gauges and the Δ line;
- **the mapper** — fills in and colours the room plan.

**Pupils sensitive to cold** do not sit by the window during the purge; the roles above are done from the far side of the room.

**If things do not go to plan**

**You have only one box.** The Δ line will show nothing and the simultaneous comparison cannot be made. **Do *Indoor Pollution* instead**, which has a complete conclusion with a single box, and come back to this app when you have two.
**You have two boxes, not six.** That is enough. Measure two locations at a time, then move to two more with ten minutes of settling. You lose simultaneity between the pairs — record the times and say so in the conclusion.
**The co-location gives a very large scatter** (over 200 ppm). Check that no box is in the sun, in a draught or beside somebody. If the scatter stays large, it *is* your threshold: the activity goes on, but the conclusions will demand larger differences.
**A box drops off WiFi mid-measurement.** Its curve stops while the others continue. Record the time and treat the interval as missing — do not interpolate it.
**You are not allowed to open the window.** Investigations 1, 2 and 4 are done in full without it. The purge can be done with the door, and the resulting $\tau$ values are still comparable between locations.
**No box will connect.** Do not stop the lesson. Use the "What the data typically looks like" table as a printed dataset: pupils are given the co-location scatter and four location values, compare each Δ with the threshold and write the three verdicts. **The whole methodological part — which is the lesson — is done without the equipment.** Measure again another day, after checking the boxes' power and the network.

## 9. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **World Health Organization** — `https://www.who.int` — search for indoor air quality and ventilation; its guidelines cover pollutants, not $CO_2$.
- **REHVA** — `https://www.rehva.eu` — the standard EN 16798-1 and European guidance on classroom ventilation; the source of professional figures for the pupils' report.
- **Sensirion** — `https://sensirion.com` — the SCD4x datasheet, if a pupil wants to see where the stated accuracy comes from.
- **STEM Learning** — `https://www.stem.org.uk` — search for air quality and survey projects for lower and upper secondary.
- **Institute of Physics** — `https://www.iop.org` — search for material on uncertainty and measurement error; it maps exactly onto the idea of a detection threshold.
- **Romanian Ministry of Education** — `https://www.edu.ro` — for the norms concerning classrooms, if the report reaches the school leadership.

## 10. Assessment

**The exit ticket**, identical to the one on the pupil sheet: *what is our detection threshold, and how do we know it?* Collected in the last two minutes, it says more about the class's understanding

than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Explorer and Scientist level the three axes look like this:

- **Measurement practice** — the pupil does the co-location before anything else, respects the settling times and the exclusion circle, and records the time, the number of people and the state of the window alongside the value.
- **Data interpretation** — the pupil compares every $\Delta$ with the co-location threshold and uses three verdicts, not two: detected, not detected, borderline.
- **Scientific reasoning** — the pupil explains why airing temporarily increases $\Delta$, frames a recommendation supported by data, and **states what the sensor cannot measure**.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

## 11. Pupil sheet

The sheet the pupils fill in is a separate document, in the folder `docs/en/student_sheets`: the file `air_quality.pdf`. It has two pages — the co-location, the threshold and the room map on the first; the simultaneous table, the comparison with the threshold and the conclusion on the second — so it is printed double-sided, one sheet per pupil. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write.

Three things to know before printing it. **The co-location threshold has a large box of its own at the head of the page**, and this is not a formality: every comparison on the sheet refers back to it, and a sheet with the threshold left blank cannot produce a valid conclusion. **The verdict column has three values, not two** — detected, not detected, borderline — because "borderline" is the correct answer whenever $\Delta$ is comparable with the threshold, and pupils need somewhere to write it. **The room map is printed, not drawn**, with the locations already marked, so that the time goes into measuring and comparing rather than into drawing. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.